

Lecture 5: Linear Regression of Time-Series Data II

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Linear Regression of Time-Series Data II

Introduction

Recap: Linear Regression of Time-Series Data

- Econometric analysis of time-series data
 - (definition, data,) identification, estimation, and inference
- We apply the classical linear regression analysis to time-series data.
 - Finite-sample properties
 - No stationarity assumption on $\{x_t\}$ and $\{y_t\}$
- TS3 (strict exogeneity) excludes any model with lagged dependent variables.
 - e.g., AR, AR-FDL, ARMA models
- TS4-2 (no serial correlation) excludes any model with an autoregressive error term.
 - e.g., ARMA models

Recap: Linear Regression of Time-Series Data

- Three messages implied by TS3 (strict exogeneity):
 - $s < t$: ε_t is uncorrelated with past x_{sk} .
 - $s = t$: ε_t is uncorrelated with contemporaneous x_{sk} .
 - $s > t$: ε_t is uncorrelated with future x_{sk} .
- Ex) an AR(1) model
 - $y_t = \alpha + \rho y_{t-1} + \varepsilon_t$
 - $y_{t+1} = \alpha + \rho \underbrace{y_t}_{x_{t+1}} + \varepsilon_{t+1}$

$\implies$ when $s = t + 1$, TS3 fails...
- TS3 is too strong for many time-series models...

Recap: Linear Regression of Time-Series Data

- Finite-sample theory:
 - Data: only a finite number of data
 - Model: strong assumptions
- Large-sample theory:
 - Data: the number of data grows to infinity
 - Model: weak assumptions

Outline

- Preliminary 1: Asymptotic Theory
- Preliminary 2: Linear regression of cross-sectional data
- Linear regression of time-series data

Linear Regression of Time-Series Data II

Preliminary 1: Asymptotic Theory

Asymptotic Theory

- A sequence of random variables $\{X_n\}$ converges in probability to a constant α : for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|X_n - \alpha| < \epsilon) = 1.$$

– Notation: $X_n \xrightarrow{p} \alpha$

- A sequence of random variables $\{X_n\}$ converges in distribution to a random variable X : for every x ,

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x).$$

– Notation: $X_n \xrightarrow{d} X$, $X_n \xrightarrow{d} F_X$

- In econometrics, estimators constitute a sequence of random variables.

Asymptotic Theory

Theorem: Law of Large Numbers

If $\{Z_n\}$ is a sequence of independent and identically distributed random variables. Then,

$$\frac{1}{N} \sum_{n=1}^N Z_n \xrightarrow{p} E[Z_n].$$

Theorem: Central Limit Theorem

If $\{Z_n\}$ is a sequence of independent and identically distributed random variables. Then,

$$\sqrt{N} \frac{1}{N} \sum_{n=1}^N (Z_n - E[Z_n]) \xrightarrow{d} \mathcal{N}(0, \Omega),$$

where $\Omega := \lim_{N \rightarrow \infty} \text{Var}(\sqrt{N} \frac{1}{N} \sum_{n=1}^N Z_n) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \text{Var}(Z_n)$.

- The case of a random vector is analogous.

Asymptotic Theory

Theorem: Continuous Mapping Theorem

Let $g(\cdot)$ be a continuous function. If $Z_n \xrightarrow{p} \alpha$, then

$$g(Z_n) \xrightarrow{p} g(\alpha).$$

- In particular, if $g(\cdot) := (\cdot)^{-1}$, then

$$Z_n^{-1} \xrightarrow{p} \alpha^{-1}.$$

Theorem: Slutsky Theorem

If $Z_n \xrightarrow{p} \alpha$ and $Y_n \xrightarrow{d} Y$, then

$$Z_n Y_n \xrightarrow{d} \alpha Y.$$

Linear Regression of Time-Series Data II

Preliminary 2: Cross-Sectional Linear Regression — Large-Sample Properties —

Cross-Sectional Linear Regression: Identification

Assumption 1 (CS1'): Linearity

A random sample $\{(y_n, x_{n1}, \dots, x_{nK}) : n = 1, \dots, N\}$ follows

$$y_n = \beta_1 x_{n1} + \beta_2 x_{n2} + \dots + \beta_K x_{nK} + \varepsilon_n,$$

where ε_n is an unobserved random error.

- In vector notation, $y_n = x_n' \beta + \varepsilon_n$, where

$$x_n = \begin{bmatrix} x_{n1} \\ \vdots \\ x_{nK} \end{bmatrix}, \quad \text{and} \quad \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}.$$

- Random sample: $\{(y_n, x_n)\}$ are independent and identically distributed across observations.
 $\implies$ so are $\{\varepsilon_n\}$.

Cross-Sectional Linear Regression: Identification

Assumption 2 (CS2'): No perfect multicollinearity

None of the explanatory variables is a perfect linear combination of the others.

Assumption 3 (CS3'): Zero conditional mean

$E[\varepsilon_n | x_n] = 0$ for all $n \in \{1, \dots, N\}$.

- CS2' is the same as CS2:
 - No redundant regressors
 - $(x_n x_n)^{-1}$ exists.
- Under CS3',
 - $E[x_n \varepsilon_n] = E[E[\varepsilon_n | x_n]] = 0$
- CS3' is weaker than CS3 (strict exogeneity).
 - Under CS3, $E[\varepsilon_i x_j] = 0$ for all $i, j \in \{1, \dots, N\}$

Cross-Sectional Linear Regression: Identification

Proposition: Identification

Under CS1'–CS3', the parameter vector β is identified.

Proof. Under CS1',

$$y_n = x_n' \beta + \varepsilon_n.$$

Premultiplying x_n and taking expectation,

$$E[x_n y_n] = E[x_n x_n'] \beta + \underbrace{E[x_n \varepsilon_n]}_{0 \text{ by CS3'}}.$$

In view of CS2', $(E[x_n x_n'])^{-1}$ exists, so that

$$\beta = (E[x_n x_n'])^{-1} E[x_n y_n].$$



Cross-Sectional Linear Regression: Estimation

- Given the identification result

$$\beta = (E[x_n x_n'])^{-1} E[x_n y_n],$$

a natural estimator for β is the sample analog:

$$\begin{aligned}\hat{\beta} &:= \left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1} \left(\frac{1}{N} \sum_{n=1}^N x_n y_n \right) = \left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1} \left(\frac{1}{N} \sum_{n=1}^N x_n (x_n' \beta + \varepsilon_n) \right) \\ &= \beta + \left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1} \left(\frac{1}{N} \sum_{n=1}^N x_n \varepsilon_n \right).\end{aligned}$$

- We want to study the stochastic properties of $\hat{\beta}$.
 - Accuracy
 - Precision
 - Distribution

Cross-Sectional Linear Regression: Estimation

Proposition: Consistency

Under CS1'–CS3', $\hat{\beta}$ is consistent for β , i.e., $\hat{\beta} \xrightarrow{p} \beta$.

Proof. Under CS1'–CS3', as $N \rightarrow \infty$,

$$\begin{aligned}\hat{\beta} &= \beta + \underbrace{\left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1}}_{\xrightarrow{p} E[x_n x_n']^{-1}} \underbrace{\left(\frac{1}{N} \sum_{n=1}^N x_n \varepsilon_n \right)}_{\xrightarrow{p} E[x_n \varepsilon_n]} \\ &\xrightarrow{p} \beta + (E[x_n x_n'])^{-1} \underbrace{E[x_n \varepsilon_n]}_{=0 \text{ by CS3'}} \\ &\xrightarrow{p} \beta.\end{aligned}$$



Cross-Sectional Linear Regression: Inference

Proposition: Asymptotic Normality

Under CS1'–CS3',

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, AVar(\hat{\beta})),$$

where $AVar(\hat{\beta}) := (E[x_n x_n'])^{-1} \Omega (E[x_n x_n'])^{-1}$ with $\Omega := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N Var(x_n \varepsilon_n)$.

Proof. By construction,

$$\begin{aligned} \sqrt{N}(\hat{\beta} - \beta) &= \sqrt{N} \left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1} \left(\frac{1}{N} \sum_{n=1}^N x_n \varepsilon_n \right) \\ &= \underbrace{\left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1}}_{\xrightarrow{p} E[x_n x_n']} \underbrace{\left(\sqrt{N} \frac{1}{N} \sum_{n=1}^N x_n \varepsilon_n \right)}_{\xrightarrow{d} \mathcal{N}(0, \Omega)} \xrightarrow{d} (E[x_n x_n'])^{-1} \mathcal{N}(0, \Omega), \end{aligned}$$

where $\Omega := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N Var(x_n \varepsilon_n)$.

Cross-Sectional Linear Regression: Inference

Proof (cont'd). Recall, in general,

$$\text{Var}(aZ) = a^2 \text{Var}(Z).$$

Hence,

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, A\text{Var}(\hat{\beta})),$$

where $A\text{Var}(\hat{\beta}) := (E[x_n x_n'])^{-1} \Omega(E[x_n x_n'])^{-1}$.

□

Cross-Sectional Linear Regression: Inference

Assumption 4-1 (CS4'-1): Homoskedasticity

$E[\varepsilon_n^2|x_n] = \sigma^2 > 0$ for all $n \in \{1, \dots, N\}$.

Assumption 4-2 (CS4'-2): No Correlation

$E[\varepsilon_i \varepsilon_j | x_i, x_j] = 0$ for all $i, j \in \{1, \dots, N\}$ such that $i \neq j$.

- Under these assumptions,

$$\begin{aligned}\Omega &:= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \text{Var}(x_n \varepsilon_n) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \left\{ E[\varepsilon_n^2 x_n x_n'] - \underbrace{E[\varepsilon_n x_n]}_0 (\underbrace{E[\varepsilon_n x_n]}_0)' \right\} \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \underbrace{E[E[\varepsilon_n^2|x_n]}_{\sigma^2} x_n x_n'} = \sigma^2 \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \underbrace{E[x_n x_n']}_{N \times E[x_n x_n']} = \sigma^2 E[x_n x_n']\end{aligned}$$

$$\implies A\text{Var}(\hat{\beta}) := (E[x_n x_n'])^{-1} \Omega (E[x_n x_n'])^{-1} = \sigma^2 (E[x_n x_n'])^{-1}.$$

Cross-Sectional Linear Regression: Inference

Proposition: Asymptotic Efficiency

Under CS1'–CS4'-2, $AVar(\hat{\beta}|X) \leq AVar(\hat{b}|X)$, where $\hat{b}$ is any other consistent and normally-distributed estimator for β .

- Among consistent and normally-distributed estimators, $\hat{\beta}$ has the lowest variance (i.e., the most precise).
 - The asymptotics counterpart of the Gauss-Markov theorem.

Cross-Sectional Linear Regression: Inference

- In practice, we do not know

$$AVar(\hat{\beta}) = \sigma^2 E[x_n x_n']^{-1}$$

- A natural estimator is a sample analog:

$$\widehat{AVar}(\hat{\beta}) = \hat{\sigma}^2 \left(\frac{1}{N} \sum_{n=1}^N x_n x_n' \right)^{-1},$$

where $\hat{\sigma}^2 := \frac{1}{N-K} \hat{\varepsilon}_n^2$ with $\hat{\varepsilon}_n := y_n - x_n' \hat{\beta}$.

- In fact,

$$\widehat{AVar}(\hat{\beta}) \xrightarrow{p} AVar(\hat{\beta}).$$

Cross-Sectional Linear Regression: Inference

- Let β_k be the k th element of β .
- We want to test

$$H_0 : \beta_k = \bar{\beta}_k$$

$$H_1 : \beta_k \neq \bar{\beta}_k,$$

where $\bar{\beta}_k$ is the hypothesized value of β_k .

- The level of significance is α .
 - Usually, α ranges from 0.01 to 0.10.

Cross-Sectional Linear Regression: Inference

Proposition: Sampling Distribution of $\hat{\beta}$

Under CS1'–CS4'-2, $t_k \xrightarrow{d} \mathcal{N}(0, 1)$, where

$$t_k := \frac{\sqrt{N}(\hat{\beta}_k - \bar{\beta}_k)}{(\widehat{AVar}(\hat{\beta}))_{kk}^{1/2}}.$$

Proof. Under H_0 , using the continuous mapping theorem and Slutsky's theorem,

$$\widehat{AVar}(\hat{\beta})^{-1/2} \sqrt{N}(\hat{\beta} - \bar{\beta}) \xrightarrow{d} AVar(\hat{\beta})^{-1/2} \mathcal{N}(0, AVar(\hat{\beta})) = \mathcal{N}(0, I_K).$$

Picking the k th element, we obtain $t_k \xrightarrow{d} \mathcal{N}(0, 1)$. □

Cross-Sectional Linear Regression: Inference

Five-Step Procedure of (Two-Sided) Hypothesis Testing

(Step 1): Determine the hypothesis $H_0 : \beta_k = \bar{\beta}_k$ vs. $H_1 : \beta_k \neq \bar{\beta}_k$.

(Step 2): Specify the level of significance α .

(Step 3): Derive the asymptotic distribution of $\hat{\beta}_k$ and construct the test statistic under H_0 :
 $t_k \xrightarrow{d} \mathcal{N}(0, 1)$.

(Step 4): Compute the critical value $t_{\alpha/2}$: $P(|t_k| < t_{\alpha/2}) = 1 - \alpha$.

(Step 5): Reject H_0 if and only if $t_k^* < -t_{\alpha/2}$ or $t_k^* > t_{\alpha/2}$.

Cross-Sectional Linear Regression: Comparison

- Compared to the finite-sample theory, the large-sample theory imposes
 - * More structures on data:
 - the number of observations grows to infinity
 - $\{(y_n, x_n)\}$ are i.i.d.
 - * Fewer structures on the model:
 - zero conditional mean independence
 - no distributional assumption on $\{\varepsilon_n\}$

Linear Regression of Time-Series Data II

Large-Sample Properties

Weak Dependence

- Independence is the key to the large-sample properties for cross-sectional linear regression.
 - LLN, CLT
- Independence is not plausible for time-series analysis.
e.g., an AR(1) process
 - $y_t = \alpha + \rho y_{t-1} + \varepsilon_t$
- We want to relax the independence requirement by considering “weak dependence.”

Weak Dependence

- Z_t and Z_{t+h} are independent if and only if

$$P(Z_t \leq z_t, Z_{t+h} \leq z_{t+h}) = P(Z_t \leq z_t)P(Z_{t+h} \leq z_{t+h}).$$

- If Z_t and Z_{t+h} are dependent,

$$P(Z_t \leq z_t, Z_{t+h} \leq z_{t+h}) \neq P(Z_t \leq z_t)P(Z_{t+h} \leq z_{t+h}).$$

That is,

$$|P(Z_t \leq z_t, Z_{t+h} \leq z_{t+h}) - P(Z_t \leq z_t)P(Z_{t+h} \leq z_{t+h})| > 0.$$

- **Idea:** Z_t and Z_{t+h} are weakly dependent if they become “almost independent” as $h \rightarrow \infty$:

$$|P(Z_t \leq z_t, Z_{t+h} \leq z_{t+h}) - P(Z_t \leq z_t)P(Z_{t+h} \leq z_{t+h})| \rightarrow 0 \text{ as } h \rightarrow \infty.$$

Weak Dependence

Definition: Weak Dependence for a Weakly Stationary Process

A weakly stationary process $\{Z_t\}$ is weakly dependent if

$$\text{Cov}(Z_t, Z_{t+h}) \rightarrow 0 \text{ as } h \rightarrow \infty.$$

- The covariance becomes negligible when the distance between the two rv's is far enough.

Weak Dependence

Theorem: Law of Large Numbers (Ergodic Theorem)

If $\{Z_t\}$ is a weakly stationary and weakly dependent stochastic process, then

$$\frac{1}{T} \sum_{t=1}^T Z_t \xrightarrow{p} E[Z_t].$$

Theorem: (Martingale) Central Limit Theorem

If $\{Z_t\}$ is a weakly stationary and weakly dependent stochastic process, then

$$\sqrt{T} \frac{1}{T} \sum_{t=1}^T Z_t \xrightarrow{d} \mathcal{N}(0, \Omega),$$

where

$$\Omega := \lim_{N \rightarrow \infty} \text{Var}\left(\sqrt{T} \frac{1}{T} \sum_{t=1}^T Z_t\right) = \lim_{N \rightarrow \infty} \frac{1}{T} \text{Var}\left(\sum_{t=1}^T Z_t\right) \neq \lim_{N \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \text{Var}(Z_t).$$

Time-Series Linear Regression: Identification

Assumption 1 (TS1'): Linearity

A jointly weakly stationary and weakly dependent stochastic process $\{(x_t, y_t)\}_{t=0}^T$ follows

$$y_t = \beta_1 x_{t1} + \beta_2 x_{t2} + \dots + \beta_K x_{tK} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ is a sequence of random errors.

- In vector notation, $y_t = x_t' \beta + \varepsilon_t$, where

$$x_t = \begin{bmatrix} x_{t1} \\ \vdots \\ x_{tK} \end{bmatrix} \quad \text{and} \quad \beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}.$$

- $\{(y_t, x_t)\}$ is weakly stationary and weakly dependent.
 $\implies$ so is $\{\varepsilon_t\}$.

Time-Series Linear Regression: Identification

- Weak stationarity and weak dependence induce constraints on the parameters β 's.
- Ex) an AR(1) process: $y_t = \alpha + \rho y_{t-1} + \varepsilon_t$
 - weak stationarity: $|\rho| < 1$
 - weak dependence: $\text{Cov}(y_t, y_{t+h}) = \rho^h \rightarrow 0$ as $h \rightarrow \infty$

Time-Series Linear Regression: Identification

Assumption 2 (TS2'): No Perfect Collinearity

None of the explanatory variables is a perfect linear combination of the others.

- This is the same as CS2'.

Time-Series Linear Regression: Identification

Assumption 3 (TS3'): Contemporaneous Exogeneity

$E[\varepsilon_t | x_t] = 0$ for all $t \in \{1, \dots, T\}$.

- The expected value of ε_t , given the contemporaneous explanatory variables, is zero.
- Under TS3',
 - $E[\varepsilon_t] = E[E[\varepsilon_t | x_t]] = 0$
 - $E[x_{tk}\varepsilon_t] = E[E[x_{tk}\varepsilon_t | x_t]] = E[x_{tk}E[\varepsilon_t | x_t]] = 0$ for $k \in \{1, \dots, K\}$.
- This assumption only says
 - $s = t$: ε_t is uncorrelated with contemporaneous x_{sk} .and does not require
 - $s < t$: ε_t is uncorrelated with past x_{sk} .
 - $s > t$: ε_t is uncorrelated with future x_{sk} .
- This assumption permits models with lagged dependent variables.

Time-Series Linear Regression: Identification

Proposition: Identification of β

Under TS1'–TS3', the parameter vector β is identified.

- This is the same as the cross-sectional case.
- β is identified as

$$\beta = (E[x_t x_t'])^{-1} E[x_t y_t].$$

Time-Series Linear Regression: Estimation

- Identified parameter: $\beta = (E[x_t x_t'])^{-1} E[x_t y_t]$
- Estimator: $\hat{\beta} := \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1} \left(\frac{1}{T} \sum_{t=1}^T x_t y_t \right)$

Proposition: Consistency

Under TS1'–TS3', $\hat{\beta}$ is consistent for β , i.e., $\hat{\beta} \xrightarrow{p} \beta$.

Time-Series Linear Regression: Inference

Proposition: Asymptotic Normality

Under TS1'–TS3',

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, AVar(\hat{\beta})),$$

where $AVar(\hat{\beta}) := (E[x_t x_t'])^{-1} \Omega (E[x_t x_t'])^{-1}$ with $\Omega := \lim_{T \rightarrow \infty} \frac{1}{T} Var(\sum_{t=1}^T x_t \varepsilon_t)$.

Time-Series Linear Regression: Inference

- In general,

$$\begin{aligned}\Omega &:= \lim_{T \rightarrow \infty} \frac{1}{T} \text{Var}\left(\sum_{t=1}^T x_t \varepsilon_t\right) \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \left\{ \sum_{t=1}^T \text{Var}(x_t \varepsilon_t) + \sum_{t=1}^T \sum_{s=1}^T \text{Cov}(x_t \varepsilon_t, x_s \varepsilon_s) \right\}\end{aligned}$$

- The easiest starting point is to assume
 1. $\text{Var}(x_t \varepsilon_t) = \sigma^2 E[x_t x_t']$ for all t
 2. $\text{Cov}(x_t \varepsilon_t, x_s \varepsilon_s) = 0$ for all t and s

Time-Series Linear Regression: Inference

Assumption 4-1 (TS4'-1): Homoskedasticity

$$E[\varepsilon_t^2 | x_t] = \sigma^2 \text{ for all } t \in \{1, \dots, T\}.$$

Assumption 4-2 (TS4'-2): No Serial Correlation

$$E[\varepsilon_t \varepsilon_s | x_t, x_s] = 0 \text{ for all } t, s \in \{1, \dots, T\} \text{ such that } t \neq s.$$

- TS4'-1 and TS4'-2 are conditional only on the contemporaneous explanatory variables.
 1. $\text{Var}(x_t \varepsilon_t) = E[\varepsilon_t^2 x_t x_t'] - E[x_t \varepsilon_t](E[x_t \varepsilon_t])' = \dots = \sigma^2 E[x_t x_t']$;
 2. $\text{Cov}(x_t \varepsilon_t, x_s \varepsilon_s) = E[\varepsilon_t \varepsilon_s x_t x_s'] - E[x_t \varepsilon_t](E[x_s \varepsilon_s])' = \dots = 0$ for all $t \neq s$.
- Under TS4'-1 and TS4'-2,

$$\begin{aligned}\Omega &= \sigma^2 E[x_t x_t'] \\ \implies \text{AVar}(\hat{\beta}) &= \sigma^2 (E[x_t x_t'])^{-1}.\end{aligned}$$

Time-Series Linear Regression: Inference

Proposition: Asymptotic Efficiency

Under TS1'–TS4'-2, $AVar(\hat{\beta}|X) \leq AVar(\hat{b}|X)$, where $\hat{b}$ is any other consistent and normally-distributed estimator for β .

- Among consistent and normally-distributed estimators, $\hat{\beta}$ has the lowest variance (i.e., the most precise).
 - The asymptotics counterpart of the Gauss-Markov theorem.

Time-Series Linear Regression: Inference

- In practice, we do not know

$$AVar(\hat{\beta}) = \sigma^2 (E[x_t x_t'])^{-1}$$

- A natural estimator is a sample analog:

$$\widehat{AVar}(\hat{\beta}) = \hat{\sigma}^2 \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1},$$

where $\hat{\sigma}^2 := \frac{1}{T-K} \hat{\varepsilon}_t^2$ with $\hat{\varepsilon}_t := y_t - x_t' \hat{\beta}$.

- In fact,

$$\widehat{AVar}(\hat{\beta}) \xrightarrow{p} AVar(\hat{\beta}).$$

Time-Series Linear Regression: Inference

Proposition: Sampling Distribution of $\hat{\beta}$

Under CS1'–CS4'-2, $t_k \xrightarrow{d} \mathcal{N}(0, 1)$, where

$$t_k := \frac{\sqrt{N}(\hat{\beta}_k - \bar{\beta}_k)}{(\widehat{AVar}(\hat{\beta}))_{kk}^{1/2}}.$$

Five-Step Procedure of Hypothesis Testing

Under TS1'–TS4'-2, the five-step procedure for hypothesis testing is valid.

Time-Series Linear Regression: Comparison

- Compared to the finite-sample theory, the large-sample theory imposes
 - * More structures on data:
 - the number of observations grows to infinity
 - $\{(y_n, x_n)\}$ are i.i.d.
 - * Fewer structures on the model:
 - zero conditional mean independence
 - no distributional assumption on $\{\varepsilon_n\}$

Time-Series Linear Regression: Example

- Examples of time-series models satisfying TS1'–TS4'-2:

- Ex1) Static model:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t$$

- Ex2) Time-series model with a time trend:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma t + \varepsilon_t$$

- Ex3) Time-series model with seasonal components:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \gamma_1 D_{Jan} + \gamma_2 D_{Feb} + \dots + \gamma_{12} D_{Dec} + \varepsilon_t,$$

- Ex4) Finite distributed lag (FDL) model of order r:

$$y_t = \beta_0 + \beta_1 z_t + \beta_2 z_{t-1} + \dots + \beta_{r+1} z_{t-r} + \varepsilon_t$$

Time-Series Linear Regression: Example

- Examples of time-series models satisfying TS1'–TS4'-2 (cont'd):
 - Ex5) Autoregressive (AR) model of order p :

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_p y_{t-p} + \varepsilon_t$$

- Ex6) Autoregressive finite distributed lag (AR-FDL) model of order p and r :

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_p y_{t-p} + \gamma_1 z_t + \gamma_2 z_{t-1} + \dots + \gamma_{r+1} z_{t-r} + \varepsilon_t$$

Time-Series Linear Regression: Example

- Examples of time-series models **NOT** satisfying TS1'–TS4'-2:
 - Ex1) Autoregressive Moving Average (ARMA) model of order p and q:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_p y_{t-p} + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \dots + \gamma_q \varepsilon_{t-q}$$

↪ TS4'-2 fails...

- Ex2) Static model with an AR(p) error term:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \gamma_1 \varepsilon_{t-1} + \dots + \gamma_p \varepsilon_{t-p} + \varepsilon_t$$

↪ TS4'-2 fails...

Summary & Preview

Summary

- We extend the linear regression framework to accommodate models with lagged dependent variables.
- We resort to the large-sample theory.
- Relative to the finite-sample theory, there are gains and losses:

Gains:

- contemporaneous exogeneity
- no distributional assumption on the error term

Losses:

- large sample
- weak stationary and weak dependence

Preview

- Our results today hinge on two simplifying assumptions:
 - Homoskedasticity (TS4'-1)
 - No serial correlation (TS4'-2)
- In the time-series context, these are not usually plausible.
 - e.g., ARMA models
- We will relax these assumptions, and study the consequences and remedies.
 - Heteroskedasticity-Autocorrelation-Robust (HAC) Estimator
 - Generalized Least Squares (GLS) Estimation
- **Next class:** March 19 (Thu), 11:00-12:15, Room 0.4, Blandijn, Campus Boekentoren